Secret Sharing

Scenarios for Secret Sharing

- Suppose you and your so-called friend do not really trust each other and are afraid that, if the other one has the map, he/she might just go alone and take everything
- Now we need a scheme that could make sure that the map is shared in a way so that no one would be left out in this trip.
- What would you suggest?

- to split the map into two pieces and make sure that both pieces are needed in order to find the island.

- You can happily go home and be assured that your friend has to go with you in order to find the island.

- This illustrates the basic concept of *secret sharing*.

We can get even more complicated.

- Maybe the general and two colonels are authorized to launch the missile, but if the general is not well, then five colonels are required to initiate a launch

- Make the launch controller so that it requires five keys. Give the general three keys and the colonels one each.

- The general together with any two colonels can launch the missile; so can the five colonels. However, a general and one colonel cannot; neither can four colonels.

A more complicated sharing scheme, called a *threshold scheme*, can do all of this and more—mathematically.

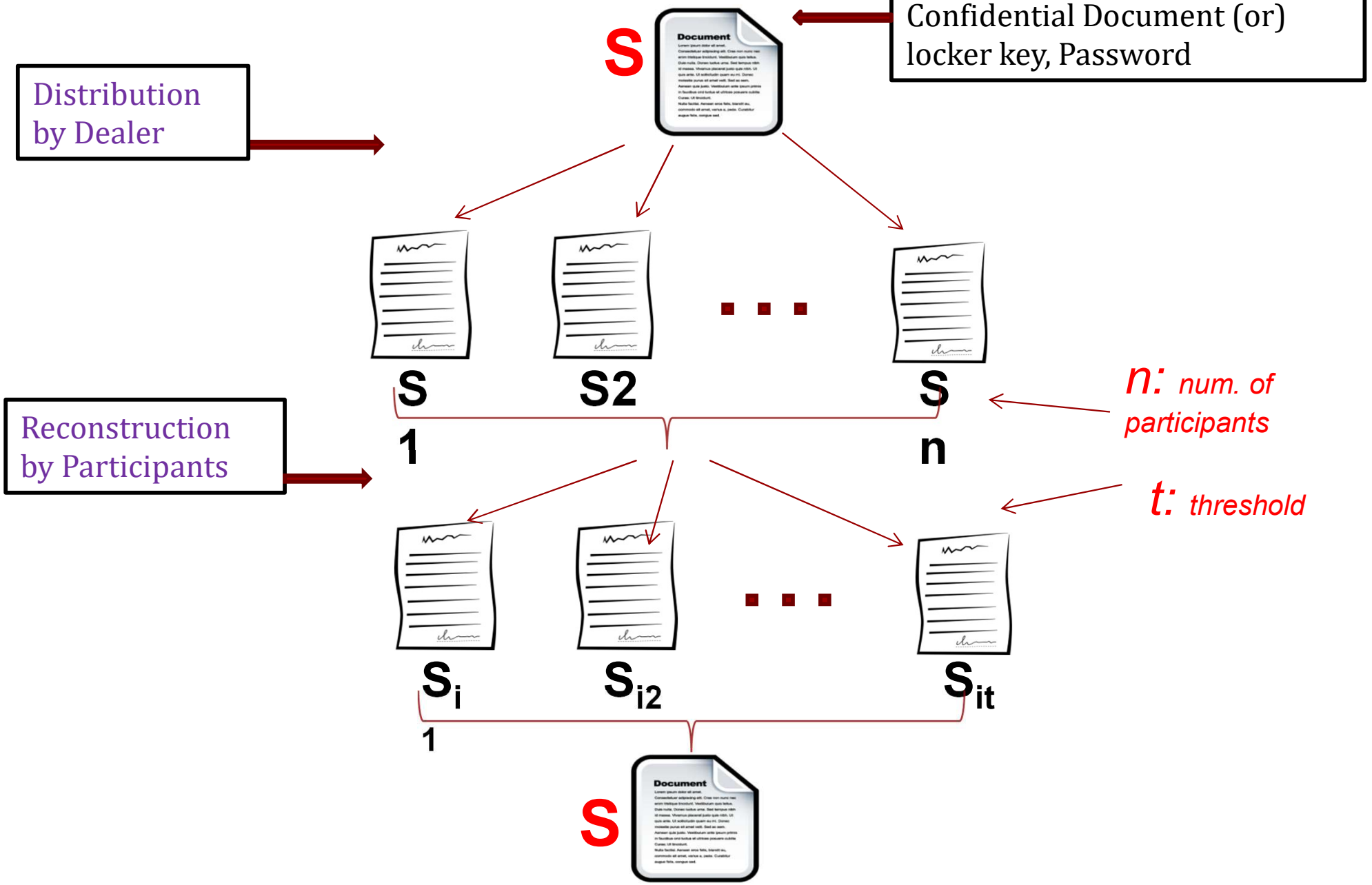
At its simplest level, you can take any message (a secret recipe, launch codes, your laundry list, etc.) and divide it into n pieces, called shares or shadows, such that any m of them can be used to reconstruct the message.

Secret Sharing

In cryptography, **secret sharing** refers to a method for distributing a *secret* amongst a group of participants, each of which is allocated a *share* of the secret.

The secret can only be reconstructed when the shares are combined together; individual shares are of no use on their own.

Secret Sharing



Threshold-Secret Sharing

Given a secret s, to be shared among n parties, that sharing should satisfy the following properties:

- Availability: greater than or equal to 't' parties can recover 's'.
- Confidentiality: less than 't' parties have no information about 's'.

Why do we want (t,n) ?

- security vs. reliability
- why not keep the key in one very safe place?
- why not have multiples copies?

- safety vs. convenience
- copies are convenient but easier to misuse
- requiring all parties is inconvenient

Applications

- Ideal for storing highly sensitive and highly important information.
- There are several applications that include:
  - Military(all Defence)
  - Banking, Mobile Environment
  - Private/GOVT. Organizations
  - Data Storage Centres
  - E-Voting
  - Cloud Computing

Shamir's Secret Sharing

- **Initialization Phase**
  - ① Dealer choose n distinct, nonzero elements, $x_1, x_2, \ldots, x_n$, of Z_P.
  - ② Public $x_1, x_2, \ldots, x_n$, and give x_i to share holder p_i.

- **Share Distribution Phase** (Suppose the secret key is K)
  - ① Dealer construct a random polynomial of degree t-1 over Z_P

$$h(x) = a_0 + a_1 x + \ldots + a_{t-1} x^{t-1}$$

    such that $a_0 = K$ (i.e. $h(0) = K$).
  - ② Dealer computes the shares $K_i = h(x_i)$ and give K_i to p_i secretly for $i = 1, 2, \ldots, n$.

Shamir's Secret Sharing (Cont.)

– **Key Recovery Phase:**

①Any t shadows can recover the $h(x)$ by pooling t points $(x_{i_1}, K_{i_1}), (x_{i_2}, K_{i_2}), .., (x_{i_t}, K_{i_t})$

②$K = h(0) = a_0$

$$h(x) = \sum_{s=1}^{t} K_s \prod_{\substack{j=1 \\ j \neq s}}^{t} \frac{(x - x_{i_j})}{(x_{i_s} - x_{i_j})} \bmod p$$

Shamir's Secret Sharing (Cont.)

- Examples: (3, 5) Secret Sharing Scheme
  - **Initialization Phase**

    ①Dealer choose 5 distinct, nonzero elements, $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$ of Z_{17}.

    ②Public $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$ and give x_i to p_i for $i = 1, 2, 3, 4, 5$.
  - **Share Distribution Phase** (Suppose $K = 13$)

    ①Dealer construct a polynomial of degree t-1 = 2 over Z_{17}

    $$h(x) = a_0 + a_1 x + a_2 x^2 = 13 + 10x + 2x^2$$

    such that $a_0 = K = 13$ (i.e. $h(0) = 13$)

Shamir's Secret Sharing (Cont.)

②Compute the shadows K_1, K_2, K_3, K_4, K_5.

$K_1 = h(1) = (2+10+13) \bmod 17 = 25 \bmod 17 = 8$
$K_2 = h(2) = (8+20+13) \bmod 17 = 41 \bmod 17 = 7$
$K_3 = h(3) = (18+30+13) \bmod 17 = 61 \bmod 17 = 10$
$K_4 = h(4) = (32+40+13) \bmod 17 = 85 \bmod 17 = 0$
$K_5 = h(5) = (50+50+13) \bmod 17 = 113 \bmod 17 = 11$

Then give K_i to p_i secretly for i=1, 2, 3, 4, 5.

Shamir's Secret Sharing (Cont.)

– **Secret Reconstruction Phase**

①Suppose the $\{p_1, p_3, p_5\}$ wants to recover the shared key $K = 13$, they pool their shares to obtains three points

$(x_1, K_1) = (1, 8)$, $(x_3, K_3) = (3, 10)$, $(x_5, K_5) = (5, 11)$.

$$h(x) = 8\frac{(x-3)(x-5)}{(1-3)(1-5)} + 10\frac{(x-1)(x-5)}{(3-1)(3-5)} + 11\frac{(x-1)(x-3)}{(5-1)(5-3)}$$
$$= [(x^2 + 9x + 15) + (6x^2 + 15x + 13) + (12x^2 + 3x + 2)] \bmod 17$$
$$= 13 + 10x + 2x^2$$

②$K = h(0) = 13$

What happens if

- Adversary modifies the contents of some of the shares
  - After Distribution Phase
  - Before Secret Reconstruction Phase



Solution: **Verifiable/Publicly verifiable secret sharing scheme**

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